Chapter 9 — IoT components and sensing

Crowdsourcing scales human labeling; the Internet of Things scales continuous measurement

Learning objectives

- By the end of this part

What the Internet of Things is

- The Internet of Things is a network of physical devices that connect to the internet and
- Smart-city systems
- Objects collect, share

Core components: devices, connectivity, processing

- At its core, IoT combines things equipped with sensors or actuators
- Devices capture measurements or perform actions
- Networks carry those readings onward
- Processing turns raw streams into decisions for users and automated systems

Example 9.12 — Wearable Health Sensors

- Example 9.12 — hands-on module
- Steps, and sleep
- These endpoints illustrate everyday sensing that produces continuous personal health
- Explore the chapter example module
- View files: `modules/chapter9/example12/`

Example 9.14 — Industrial Machine Monitoring

- Example 9.14 — hands-on module
- Predict maintenance needs
- Industrial IoT therefore links sensing to operational reliability rather than only
- Explore the chapter example module
- View files: `modules/chapter9/example14/`

Connectivity options for IoT transfer

- Devices need reliable links to gateways or cloud services
- Common options include Wi-Fi for home and office devices
- Choice depends on power, range, and bandwidth

Example 9.16 — Soil Sensors for Continuous Agriculture Data

- Example 9.16 — hands-on module
- Example 9.16 highlights soil sensors that support continuous agriculture data
- Farmers collect moisture, temperature
- Explore the chapter example module
- View files: `modules/chapter9/example16/`

Takeaways

- IoT collection rests on sensing devices, connectivity, and edge or cloud processing
- Wearables and industrial monitors show personal and plant-floor endpoints
- Connectivity choices and soil networks illustrate how continuous streams replace sparse

Next

- Complete the quiz for this part
- Local decisions such as thermostats, traffic lights, and logistics rerouting